ModelIt!

Student Activity Pack

Can Plants Survive Without Bees?

The Amazing World of Pollination

Grade 3 | NGSS 3-LS2-1 | Systems and System Models

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Have you ever watched a bee visit a flower? What was the bee doing?**

2. **Where do you think fruits like apples and strawberries come from? How do they grow?**

3. **Draw a bee visiting a flower and show what you think is happening.**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Pollinator Visits | Pollination Success | Fruit Production

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

ModelIt!

Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: No Pollinators

Pollinator Visits set to zero

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Many Pollinators

Pollinator Visits set to maximum

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Some Pollinators

Pollinator Visits set to medium

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

ModelIt!

Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Flower Diversity — How many different types of flowers are available — more variety attracts more types of pollinators

Pesticide Use — Chemicals sprayed on crops to kill pests — some pesticides accidentally harm pollinators too

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

If all the bees disappeared tomorrow, would our food still grow? Use evidence from your simulation to support your answers.

1 Why do we say bees and flowers NEED each other — what would happen if one disappeared?

2 How did your model help you understand why protecting pollinators is so important for our food?

8 STEM Challenge: Build a Pollinator Garden Design

ModelIt!

YOUR ENGINEERING MISSION

Design a garden that attracts the most pollinators possible, using the right flowers, colors, and layouts to create a pollinator paradise.

THE SCENARIO

Your school wants to help bees and butterflies by planting a pollinator garden. Your team must design the garden to attract the most pollinators. Use what you learned about pollination to choose the best flowers, arrangement, and features!

Engineering Design Thinking

1. WHICH FLOWERS WILL YOU PLANT — WHAT COLORS, SHAPES, AND SCENTS ATTRACT POLLINATORS?
2. HOW WILL YOU ARRANGE THE FLOWERS — IN ROWS, CLUSTERS, OR SCATTERED PATCHES?
3. WILL YOU INCLUDE A WATER SOURCE FOR POLLINATORS LIKE A SHALLOW DISH WITH PEBBLES?
4. HOW WILL YOU MAKE SURE SOMETHING IS ALWAYS BLOOMING FROM SPRING THROUGH FALL?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Entomologists study insects including bees and butterflies, learning how they live, pollinate, and affect ecosystems. They earn $45,000-$85,000/year.